REVIEW

Glioblastoma: Insights into Pathogenesis, Molecular Landscape, and Emerging Therapeutic Frontiers Including CRISPR-Based Strategies

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Abstract

Glioblastoma (GBM), the most intense form of primary brain tumor, continues to pose significant therapeutic challenges due to its genetic heterogeneity, pervasive nature, and resistance to standard treatments. The advent of CRISPR-Cas gene-editing technology has revolutionized precision oncology, offering powerful new strategies to target key oncogenic drivers and redefine the molecular framework of GBM. This review consolidates recent progress in CRISPR-based approaches aimed at silencing fundamental oncogenes implicated in GBM pathogenesis, while also exploring the obstacles of delivering CRISPR components effectively across the blood-brain barrier (BBB).

We examine up-to-date delivery systems designed to enhance central nervous system (CNS) penetration and tumor-specific targeting. Key safety concerns, including immune activation and off-target effects, are critically assessed to guide the development of clinically feasible interventions. Additionally, we highlight promising findings from preclinical models demonstrating CRISPR's potential to inhibit tumor growth, increase chemosensitivity, and modulate the immunosuppressive tumor microenvironment. Finally, the review explores emerging clinical trials investigating CRISPR-edited CAR-T cells and in vivo gene editing, marking pivotal advances towards clinical translation.

Keywords: Glioblastoma, CRISPR-Cas9, gene editing, blood-brain barrier, CAR-T cells, tumor microenvironment, oncogene silencing, precision oncology

Introduction

Glioblastoma (GB), *IDH* wild-type (wt), is the most prevalent and aggressive form of glial tumors, representing nearly 50% of primary malignant tumors in the central nervous system (CNS) [1, 2]. It is classified as a grade 4 tumor according to the world Health Organization (WHO) Classification of CNS tumors. GB belongs to the "Adult type diffuse glioma" group, which also includes astrocytoma, IDH-mutant (WHO grade 2, 3, or 4), and Oligodendroglioma, IDH-mutant with 1p/19q co-deletion (WHO grade 2 or 3) [1].

Unlike astrocytoma IDH-mutant WHO grade 4, IDH-wildtype glioblastoma (GB IDHwt) develops suddenly, without any signal, and progresses rapidly after a brief clinical history. Despite intensive treatment, the diagnosis of GB, IDHwt remains poor, with an average survival time of 8 months and a 5 year relative survival rate of just 5.5% [2].

Despite ongoing research, the exact mechanisms driving glioblastoma development remain unclear. However, all-inclusive molecular analysis of GB over the past 20 years has shed light on the underlying processes of glioma formation [3, 4]. Studies have thoroughly characterized the genomic and epigenomic features of GB, leading to the classification of three molecular subtypes – preneural, classical, and mesenchymal – based on gene expression profiles [3, 5, 6, 7]. Although this classification has not yet significantly impacted clinical treatment, it has revealed new pathways for research into GB pathogenesis, including the discovery of glioma-initiating cells, and has highlighted the importance of addressing intratumoral heterogeneity and differentiation plasticity in GB.

Recent research has proposed the idea that GB may arise from neural stem cells (NSC) and glial precursor cells, such as oligodendrocyte and astrocytic precursor cells [9, 10]. Moreover, it has been shown that the originating cell lineage plays an essential role in tumor molecular stratification, regardless of the driver mutation that it initially carries. [11, 12]. The cellular origin of mesenchymal glioblastoma remains ambiguous, while it has been proposed that proneural and classical GB was initiated by glial or neuronal progenitor cells .Studies

have described a potential proneural to mesenchymal transition (PMT) that may illustrate the transcriptomic plasticity of GB upon treatment or recurrence. In the latest findings, the potential cells of origin in mesenchymal GB have originated by neural crest (NC)-derived cells. [13]. Herein, we discuss perivascular mesenchymal stromal cells (pMSC), also referred to as vascular fibroblasts (vFB), which originate from the NC, as potential candidates for the initiation of GB and their role in GB development [14].

Revolutionary shift and viewpoints on the categorization of glioblastoma, Glioma was initially described using post-mortem histological examination by Rudolf Virchow in 1865 where he found out that malignant tumors were derived from glial cells. [60] The infiltrative nature of gliomas was later confirmed by Heinrich Sroebe , He further explained the intratumoural heterogeneity observed by using the concept of histological subtypes .[60] .This results in one of the most significant discoveries which highlights the importance of multifocal examination of brain tumours and also helps in better understanding the biological mechanism of glioblastoma. At the molecular level , we can characterize the intra- and intertumoural heterogeneity by using advanced technology .

In 1884 , Initial documented cranial operation for the resection of a brain tumour was performed by Rickman Godlee .In the 1900s, the neurosurgical field progressed significantly, starting with Harvey Cushing's revolutionary surgical technique that incorporated subtemporal decompression to control intracranial pressure.

By 1910, Cushing Had introduced this approach as the first major surgical progress to substantially reduce operative mortality [61,/62].Classification of brain tumors began in 1925 after Globus and Strauss proposed the name 'spongioblastoma multiforme' for the unique clinical and histological characteristics of a subtype of malignant glioma.[63] .In 1926 , Bailey and Cushing challenged this terminology and suggested a classification system for brain tumours based on their cell-of-origin, which leads to the suggestion of the originally used term, 'glioblastoma multiforme'.[64] The early 1900s saw ongoing debate over the surgical removal of malignant brain tumors along with the argument which appropriate terminology should be used . This was primarily due to the risks and complications involved in locating the lesions as well as the limited improvements in clinical outcomes. The introduction of ventriculography, the Potter-Bucky X-ray Grid, and angiography improved localisation of tumours and visualization of vasculature, which rapidly advancing surgical practices.[65, 66] However, it wasn't until the late 1950s that two milestone clinical studies demonstrated the relevance of the degree of surgical resection onpatient survival and introduced the beneficial role of adjuvant radio therapy (RT) [67, 68]. In 1997 the significant breakthrough in glioblastoma treatment emerge with the discovery of the oral alkylating agent temozolomide.(TMZ).[69] Over the succeeding decade, work by Roger Stupp and colleagues showed that concomitant TMZ and RT significantly improved OS and progression-free survival(PFS) compared with RT alone, but only for a short period of time [79,70]. More recently, deeper understanding of glioblastoma pathobiology was possible due to the advanced technology especially continued evolution of genetic sequencing capabilities. In recent years, the knowledge regarding the cell-of-origin of glioblastoma tumors has improved rapidly due to the spike of 'big data', which results in the way we think about the classification of glioblastoma has changed. Over the past 15 years,tumor cell states in glioblastoma have been intensively investigated. While initial bulk sequencing studies identified the most highly prevalent glioblastoma mutations,[71] research focus has since evolved into developing a framework for glioblastoma subtypes. There are three subtypes of glioblastoma which have been identified based on gene expression and copy number changes are proneural, mesenchymal, and classical. [72,73] . Distinct alterations associated with each subtype such as neurofibromatosis type 1 (NF1) in the mesenchymal subtype, epidermal growth factor receptor (EGFR) in the classical subtype and platelet-derived growth factor receptor α (PDGFRA) in the proneural subtype, have provided clues to their cell-of-origin. For example, the proneural subtype is enriched for the oligodendrocyte development gene signature. Therefore, this subtype may arise from a progenitor or neural stem cell in the oligodendrocyte lineage, such as PDGFRA-positive cells in the sub-ventricular zone, one of the main neurogenic regions in the adult brain.[73,75] Bulk transcriptomics have enabled the prediction of intra-tumoural heterogeneity, allowing researchers to deconvolute an individual tumour to estimate the mixture of glioblastoma subtypes. This method also highlights any differences in cellular composition between the tumour core and the tumour periphery.[76]

Recent studies have transfered from conventional bulk transcriptomic analysis to single cell RNA sequencing (scRNA seq) analyses which allow the transcriptional characterization of thousands of individual cells from a single biological sample [77,78]. This technology has been important in redefining subtypes in respect to the normal brain. Recently, using direct parallels with normal developmental cell states an scRNA seq study compared the human foetal brain with glioblastoma samples to define tumour subtypes .[78] These tumour cell states include neural progenitor cell-like (NPC-like), astrocyte-like (AC-like),oligodendrocytic precursor cell-like (OPC-like), each of which closely parallel normal cell states, and mesenchymal-like (MESlike), which has similarities to radial glia, but is more distantly associated with normal glial cells.[77,78] These studies provide the strongest evidence in identifying the origin of glial cells of glioblastoma in patient samples. Furthermore, it has also been shown by scRNA seq that the composition of tumor is not only fluctuating across patients but it is immensely heterogeneous within a single tumour.[74,77,78]. Recent longitudinal studies following treatment exposure have also shown that the proportion of tumour cells within each subtype is plastic and

changes as a result of therapeutic insult.[79] Specifically, the prevalence of the mesenchymal subtype may increase due to a higher portion of cycling mesenchymal cells following therapy.[79] Unfortunately, despite advanced technology and a deeper understanding of glioblastoma subtypes over the the past century has not led to improved patient outcomes.

The average survival rate of glioblastoma patients in the 19th century was remarkably similar to the outcomes observed in patients today [67,68]. Therefore, to enhance the outcomes and quality of life of glioblastoma patients, we urgently need more effective targeted therapies.

General characteristics

Epidemiology

In adults, the most common malignant CNS tumor is GB, *IDH*wt . It accounts for approximately 15% of all intracranial neoplasms and almost 50% of all malignant CNS tumors. It predominately affects older adults, with the highest incidence occurring in individuals between the ages of 55 and 85.(median age of 64 years). In the United States of America, GB, *IDH*wt is more common in males compared to females (M: F ratio of 1.58: 1) [1, 2]. Till date, ionizing radiation exposed to the head and neck is the only validated risk factor [15, 16]. In contrast, decreased risk has been observed among individuals with a history of allergies or atopic diseases [16]. Despite a multimodal therapeutic approach that includes surgery, radiotherapy, and chemotherapy, prognosis remains poor, with a 5-year survival rate of 5.5% [2, 17, 18].

Definition of GB

GB, *IDH*wt is a rapidly growing brain tumor which is filled with tiny abnormal blood vessels and dead tissues . The previous term “GB multiforme” exhibits the internal structure of GB which highlights that brain tumour can look very different between patients and even within the same tumour . Cellular pleomorphism includes small, undifferentiated cells, spindled, cell containing fat , granular, epithelioid, and/or giant cells. Secondary structures of Scherer describes the different routes which is taken by glioma cells to invade the brain: 1) through the white matter pathway, 2) the vasculature (perivascular satellitosis), 3) the leptomeningeal space and 4) into the brain tissue (parenchyma).

By definition, GB, *IDH*wt does not have any specific mutations in the IDH1 gene at codon 132 or IDH2 gene at codon 172. At molecular level ,the glioblastoma can be diagnosed by finding *TERT* promoter mutations, *EGFR* gene amplification, and a gain of chromosome 7or loss of chromosome 10 [1].

GB Molecular pathways

Molecular Alterations in PI3K and MAPK Pathways in IDHwt GBM

The receptor tyrosine kinase (RTK) activates the PI3K and MAPK pathways, which regulate many cellular processes, including cell proliferation. In approximately 90%of GB, at least one activating alteration in the PI3K pathway is observed, including alterations of RTK genes, PI3K genes, and *PTEN* [3, 19]. The most common alteration of GB is RTK genes, involving *EGFR* (60%), *PDGFRA* (10–15%), *MET* (2–5%), or *FGFR3* (~ 3%) [3, 5, 20]. PI3-kinase mutations are found in about 25% of GB [3]. In addition, the *NF1* gene, which encodes neurofibromin that functions as a negative regulator of RAS signaling, is deleted or mutated in 10% of cases [3].

Alterations in the p53 Pathway in Glioblastoma

In a large variety of tumors, there is an alteration of p53 pathway, including GB. Indeed, up to 90% of GB have an altered p53 signaling pathway, with mutation or deletion of *TP53* in 20–25% of cases [3, 19]. Additionally, about 15% of GB exhibits an amplification of *MDM2* or *MDM4*, thus inhibiting p53 [3]. Furthermore, approximately 60% of GB cases involve the inhibition of MDM2 due to the homozygous deletion of the CDKN2A locus, which encodes the p14ARF protein, resulting in an inactivation of the p53 pathway and the pRB pathway. [3].

CDK4/6–CDKN2A/B–RB1 cell-cycle pathway

The pRB pathway represents a critical cell cycle checkpoint, suppressing cell cycle entry (Fig. 1). The progression from G1 to S phase of the cell cycle is due to the downstream inhibition of pRB which is suppressed by CDK4 and CDK6. P16, encoded by *CDKN2A*, inhibits CDK4 and CDK6. Approximately 80% of glioblastomas (GB) exhibit alterations in the pRB signaling pathway. These alterations include deletions of *CDKN2A*, amplifications of *CDK4*/*CDK6*, and inactivating alterations of *RB1* [3, 19].

The origin of GB

The precise cellular origin of GB has remained unidentified. The potential candidates for initiating GB are several cell types within CNS, including neural precursor cells (NPC), oligodendrocyte precursor cells (OPC), and astrocytic precursor cells (APC (Fig. 2) [9, 21, 22]. Modern research shows the cell lineage plays a vital role in determining the molecular subtype of GB. Indeed, the introduction of identical driver mutations in different precursor cells leads to the development of distinct molecular subtypes [11, 12].

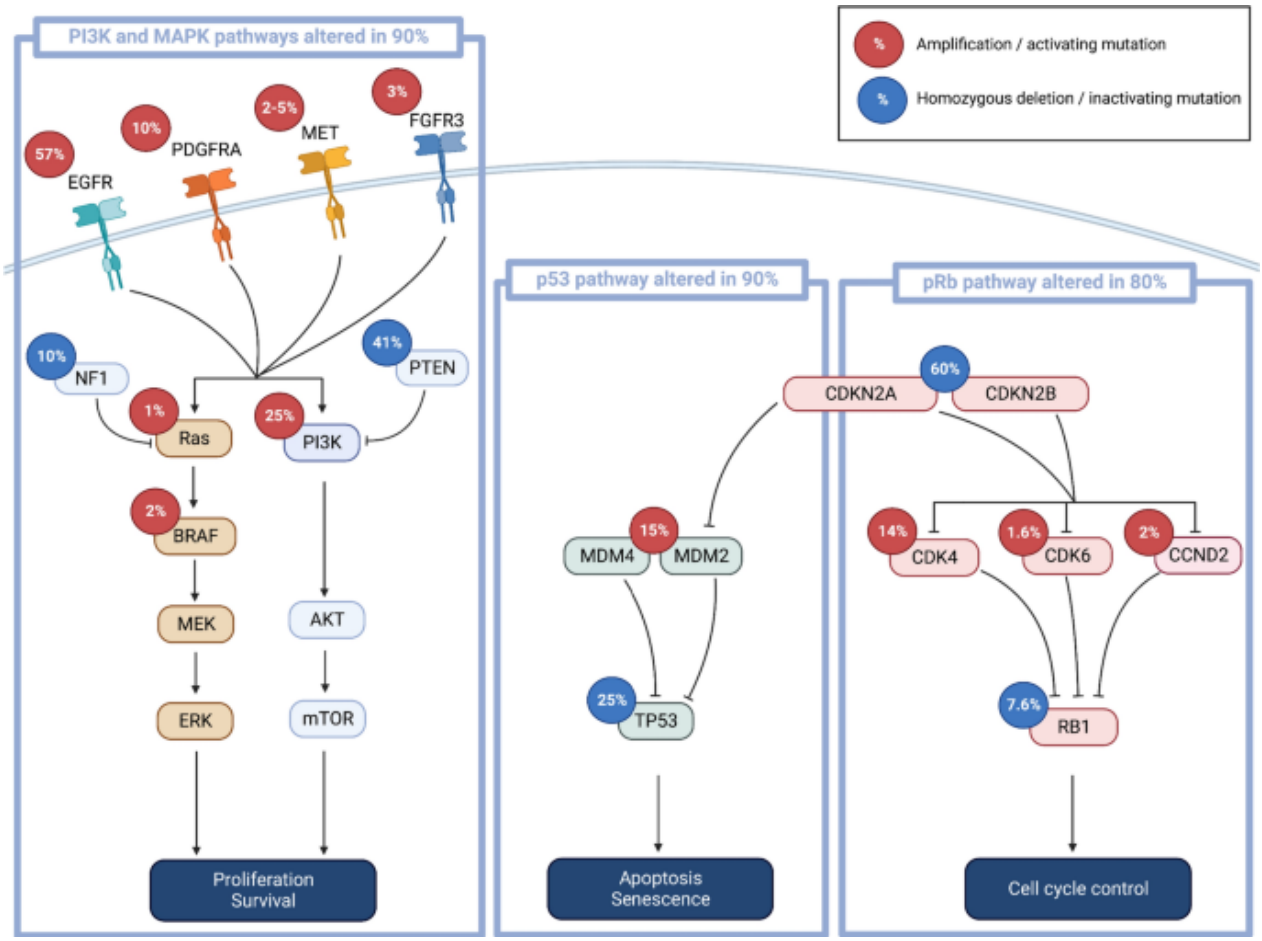


Fig. 1 Signaling pathways elaborate in GB. Variation rates are brief for PI3K/MAPK, p53 and pRb regulatory

Research has shown that neural stem cells (NSC) which is located in the subventricular zone carry the driver mutations responsible for GB, suggesting them as a potential cell of origin [10, 23, 24]. Further supporting this notion, single-cell RNA-sequencing (scRNAseq) studies have identified profiles resembling NPC, OPC, and APC, providing evidence for a neuronal/glial origin of GB [8, 25]. However, the cellular origin of GB having characteristics of mesenchyma remains unclear, despite the well-described mesenchymal transcriptomic profile (Fig. 2).

CD133-positive glioma stem cells (GSCs) have been identified within glioblastoma tumors, lending support to the hierarchical tumor development model. These GSC displays Exceptional proliferation capacity, self-renewal abilities, and differentiation potential [26,27]. They are characterized by the expression of CD133, OCT4, CD44, nestin, and SOX2, although a specific marker exclusive to GSC has not yet been identified [26, 28, 29]. GSC are described as slow-dividing or quiescent cells that are found within protective microenvironments known as GSC niches, which takes part in intratumoral heterogeneity and therapy resistance [26, 30, 31]. GSC are often located in the perivascular niche, where they interacting with endothelial cells .(Fig. 3) [32].

GB immune microenvironment

GB is a highly complex tissue that is made of tumor cells and their surrounding microenvironment which supports tumor growth by forming a supporting niche. The tumor microenvironment (TME) consists of various components which including immune cells, vascular cells, glial and neuronal cells, stem cells, soluble factors, signaling molecules, and an extracellular matrix. It is an evolving and interactive environment that plays a vital role in the production of tumour through interactive communication with tumor cells [41]. GB TME is compartmentalized in tumor niches which are distinct regions that facilitate close interactions between tumor cells and various host cell populations(Fig. 3).

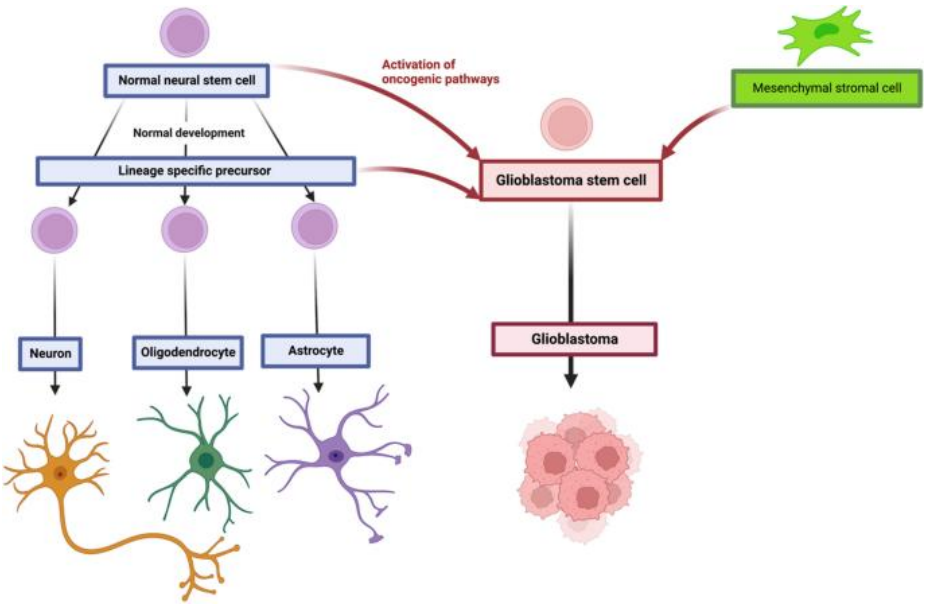


Fig. 2 The origin of glioblastoma. During normal embryonic development and in the adult brain, normal neural stem cells generate glial and neuronal cells. Glioblastoma stem cells may arise from neural stem cells and/or glial precursor cells through the activation of oncogenic pathways. They may also originate from neural crest (NC)-derived, pMSC. During development, the NC arises from the neural tube and its component cells migrate and invade virtually all tissues, giving rise to numerous differentiated cells, such as pMSC, melanocytes, chondrocytes, peripheral neuronal and glial cells, thyroid C cells, and adrenergic cells (created with Biorender.com)

CRISPR Therapy

CRISPR-Based Gene Editing in Tumor Therapy

The use of CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) in tumor therapy can redefine the treatment of various diseases [81]. The technology of CRISPR enables precise and efficient manipulation of the genome, especially its application in tumor research, offering the potential to target specific genetic mutations involving the growth and progression of tumors [82]. Over the past few years, there is extensive research on the use of CRISPR-based gene editing in tumor therapy, with several preclinical studies and clinical trials demonstrating promising results [83]. Since 2012, the discovery of CRISPR has established a new era in the field of genome editing [84]. The evolution of CRISPR tools used for exploring tumor biology is shown in Fig.4.

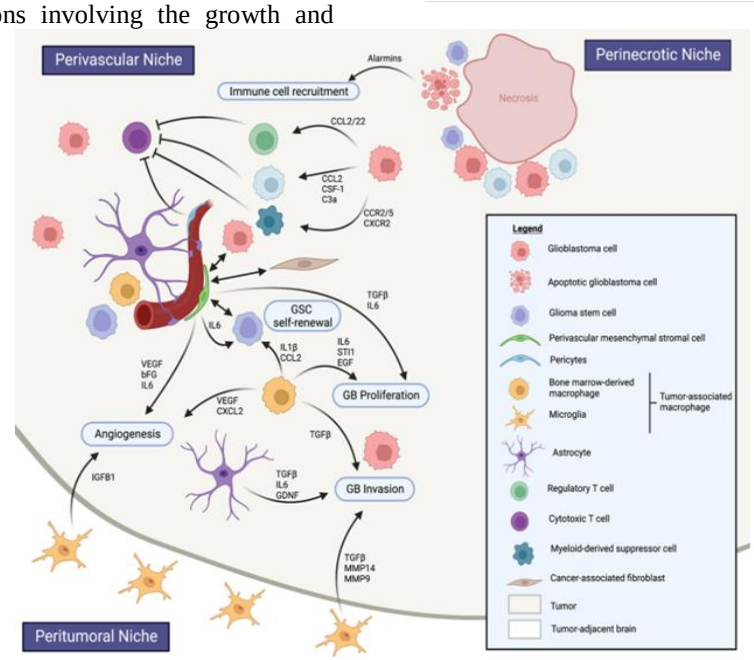


Fig. 3 Glioblastoma tumor micro-environment. GB TME is compartmentalized in perivascular, perinecrotic and peritumoral niches. Tumor-associated macrophages (Bone marrow-derived macrophages and microglia) and mesenchymal stromal cells play key roles in supporting GB proliferation, invasion and angiogenesis (created with Biorender.com)

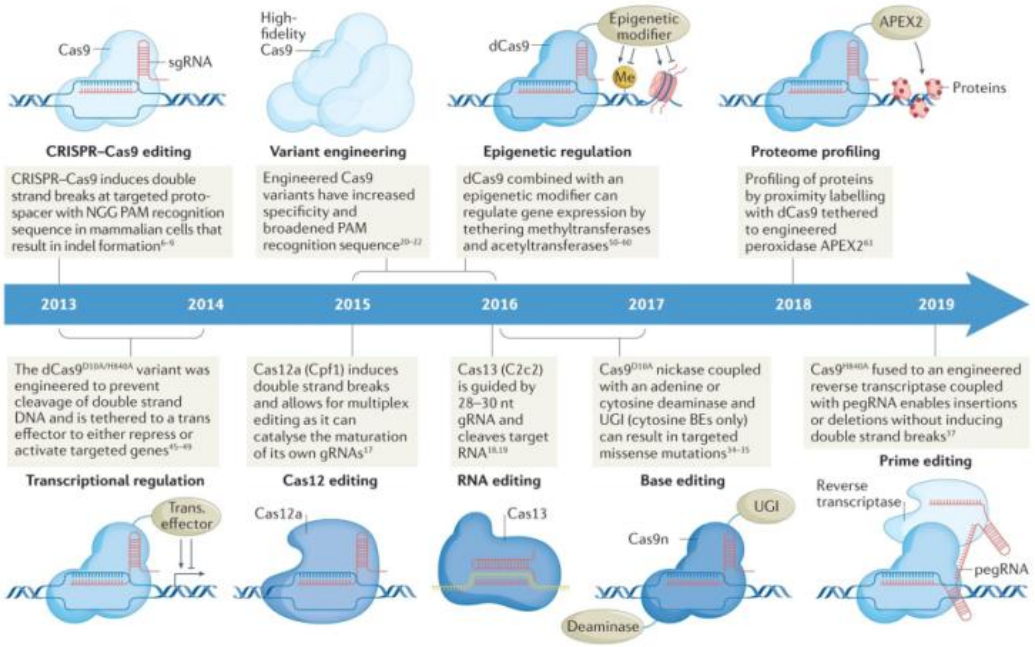


Fig. 4 The evolution of CRISPR tools that have been harnessed in the investigation of tumor biology. Since the inception of CRISPR-associated 9 (Cas9) gene editing in mammalian cells, there has been a rapid expansion in the field of CRISPR technology. This expansion has led to the development of various specialized CRISPR variants designed to tackle specific challenges. Scientists have created these variants through deliberate design and evolutionary processes, resulting in improved flexibility in recognizing protospacer adjacent motifs (PAMs) and increased precision in target selection. Additionally, they've harnessed naturally occurring variants from different bacterial species, like Cas12a (Cpf1) and Cas13, for effective combinatorial knockout (KO) and precise RNA targeting, respectively. To broaden the range of CRISPR applications, researchers have combined transcriptional effectors with catalytically inactive Cas9 (dCas9), allowing precise targeting of the transcriptome and epigenome. Furthermore, CRISPR base editing has enabled the introduction of specific transition mutations using a Cas9 nickase (Cas9n) fused with adenine or cytosine deaminase. In the case of cytosine base editing enzymes (BEs), they use a uracil glycosylase inhibitor (UGI) to prevent base excision repair and promote C >T transition mutations. A significant advancement known as prime editing has emerged, which involves fusing a dCas9 with a reverse transcriptase, enabling the engineering of various mutation types, such as missense mutations, insertions, and deletions. This is guided by a sequence template and an extended prime editing guide RNA (pegRNA). Additionally, to facilitate unbiased proteome mapping, researchers have employed engineered ascorbate peroxidase (APEX2) tethered to dCas9, enabling targeted biotinylation at specific genomic locations.[91]

CRISPR-associated enzymes, such as Cas9, can be instructed to target specific DNA sequences, using guide RNAs, allowing them to cut, modify, or delete genes in a precise manner [84]. This technology has been used in a wide range of applications, including basic research, gene therapy, and agriculture [81]. However, its possible implementation in tumor research has targeted particular interest due to the ability to target the genetic mutations that drive the growth and spread of tumors [83].

Pathogenesis

The most common primitive brain tumor in adults is IDH-wild-type glioblastoma (GBM) [92]. Over the last twenty years, the outlook for patients remains poor, and the standard treatment approach—consisting surgical removal of the tumor, radiation therapy, and alkylating chemotherapy has seen little advancement [93].

In other gliomatypes, some oncogenic mutations have proven to be actionable clinical targets. Promising clinical results of IDH-inhibitors for IDH-mutant astrocytoma, BRAF inhibitors for BRAF-mutant glioma, and MEK inhibitors for pediatric low-grade glioma has shown the opportunity of targeted therapy for gliomas, but unfortunately not yet in GBM [95,96]. Identification of targetable oncogenes and development of effective targeted therapies for GBM remains an area of active investigation.

Role of Neurofibromin 1 (NF1) in GBM

A prospective target for intervention in the MAPK pathway is Neurofibromin 1 *(NF1).* NF1 is a tumor suppressor that functions as a activating protein RAS GTPase, negatively regulating RAS/MAPK signaling by promoting the inactive GDP-bound form of RAS. Neurofibromatosis type 1 (NF-1) is a tumor predisposition syndrome which is caused by Germline mutations in *NF1* linked with benign and malignant tumors predominantly present in the peripheral and central nervous system. In many sporadic tumors including melanoma ,lung tumor, ovarian tumor, causes somatic mutations which are implicated by *NF1* [97]. In GBM, *NF1* loss is more common at recurrence and is linked with the mesenchymal subgroup, which has the worst clinical results [98]. The inactivation of NF1 has been shown to occur through multiple mechanisms. In addition to mutations leading to gene inactivation, the inactivation of NF1 occurs through other mechanisms such as aberrant splicing, post-transcriptional silencing, and proteasomal degradation [99,100].

Challenges in Detecting Functional NF1 Loss

There are several challenges that occur due to functional loss of NF1 using next-generation

sequencing (NGS):

- NF1 is a large gene (~350 kbp), making full sequencing and difficult to identify reliably.
- The functional effect of many single nucleotide variants (SNVs) is vivid.
- In standard clinical genetic sequencing non-coding or regulatory mutations may be missed.
- NGS cannot detect post-transcriptional silencing or proteasomal degradation, which may result in functional loss.

CRISPR-based strategies for tumor therapy

CRISPR-based gene editing holds the ability to bring change in tumor treatment by enabling genome modification to target the specific genetic mutations responsible for tumor development and progression [107]. For tumor therapy, different CRISPR-based strategies have been proposed [84]. One of the strategies that is used is the inactivation of genes that drive tumor growth. For instance, a way to suppress tumor growth is by using CRISPR to inactivate the oncogene MYC has been suggested. The MYC gene is known to be overactive in many types of tumor, and its inactivation could potentially inhibit or halt the progression of the disease [85].

Another strategy is the stimulation of immune response to tumor cells. For instance, researchers have used CRISPR-based gene editing to knockout or suppress the expression of the PD-1 protein on T cells, which helps to improve their ability to target and kill tumor cells [86]. Additionally, another application of CRISPR-based gene editing is that they can be used to repair genetic mutations that cause tumor, such as in the case of genetic tumor caused by BRCA1 and BRCA2 mutations [87]. Moreover, CRISPR-based gene editing can also be used in immunotherapeutic strategies for tumor treatment. Preclinical studies and clinical trials have been conducted using these strategies, and they have demonstrated promising results [83]. For example, elevating the expression of PD-1 on T cells has been shown to enhance the ability of these cells to target and kill tumor cells in animal models [81]. Table 1 outlines several CRISPR-based strategies for tumor therapy.

Figure 5 illustrate the step-by-step process of CRISPR screening, starting with the identification of specific gene targets. Subsequently, it highlightens the design and construction of a CRISPR guide RNA libraries, important for accurate genomic targeting. Following this, the delivery of CRISPR components into target cells is illustrated, demonstrating the methods used for gene editing across various cell types.

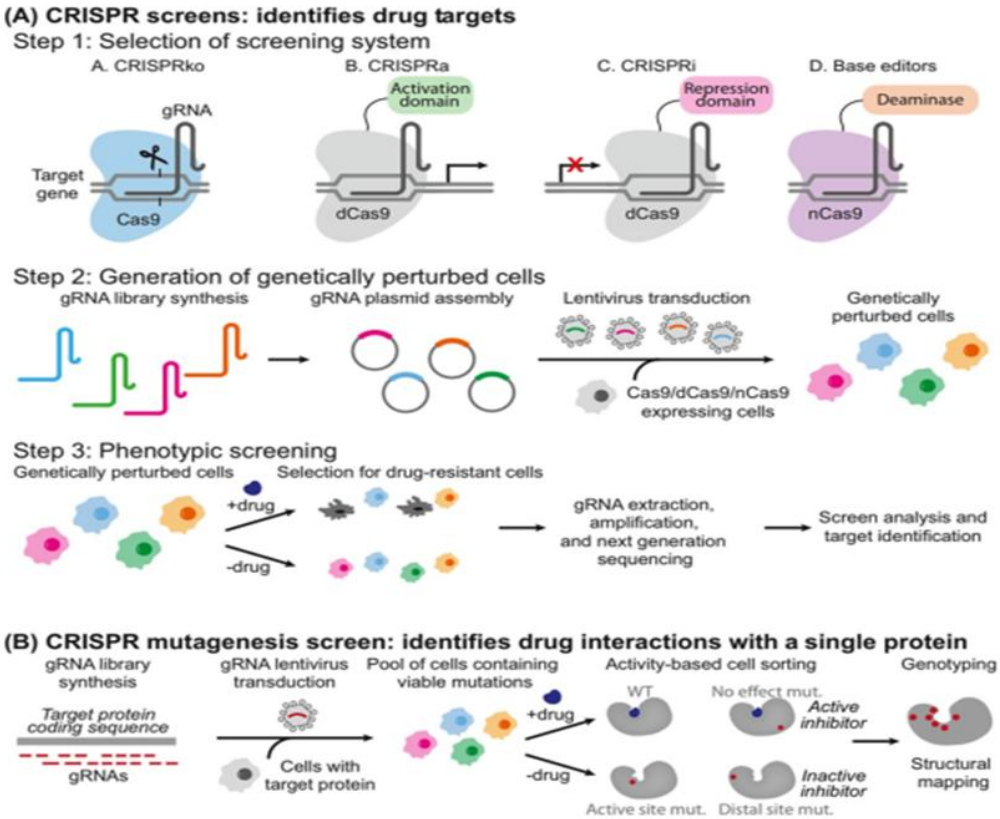


Fig. 5 Different workflows used in CRISPR screening and mutagenesis. The CRISPR screening procedures commence by selecting the appropriate screening system, offering various options: A CRISPRko, where Cas9 is employed to disrupt genes, resulting in the generation of premature stop codons or frameshift mutations; CRISPRa, involving the attachment of activation domains (e.g., VPR, VP64) to dCas9, resulting in enhanced transcription of target genes; CRISPRi, on the contrary, employs repression domains (e.g., KRAB) tethered to dCas9, leading to a reduction in the transcription of target genes; Base editing screen, which uses a base editor (e.g., cytosine

deaminase or adenine deaminase) with or without a uracil DNA glycosylase inhibitor to induce mutations without causing double-strand breaks. Once the suitable CRISPR screening method is chosen, the gRNA library is introduced into cells, creating a genetically altered cell population. These cells are exposed to drugs to select for drug-resistant populations. Subsequently, the gRNAs are extracted from the cells, amplified via PCR, and their target genes are determined using next-generation sequencing. B On the other hand, CRISPR mutagenesis screening begins with a gRNA library designed to induce in-frame mutations in the target protein coding sequence. After transducing the cells with the gRNA library, viable cells with protein variants are subjected to drug treatment, both with and without the drug. Activity-based cell sorting is used to enrich cells carrying mutations that make the drug ineffective, thereby identifying drug-resistant cells. Finally, the enriched cells are genotyped using deep sequencing to analyze structural changes
and detect any escape mutants [109] .

Figure 6 illustrates the various mechanisms of gene editing. Several different CRISPR-based strategies have been proposed for tumor therapy, each with their own advantages and limitations [108]. Table 1 outlines several
CRISPR-based strategies for tumor therapy.

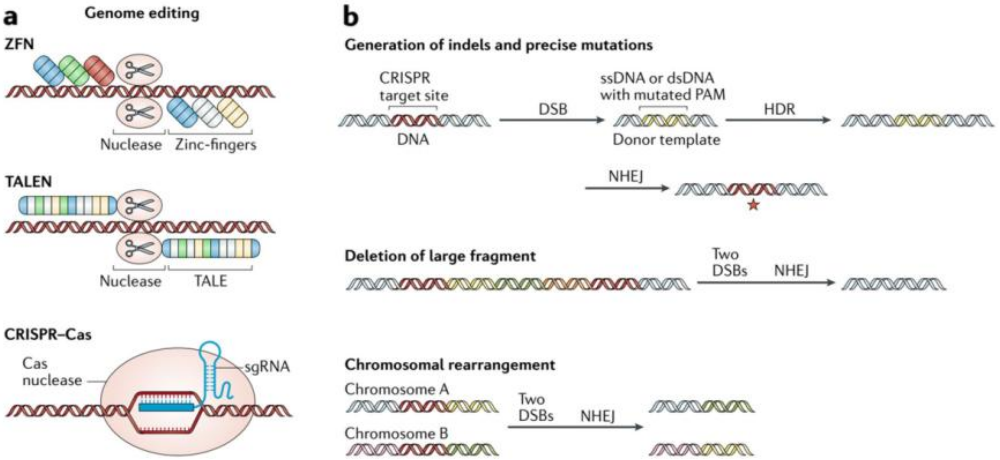


Fig. 6 The various mechanisms employed for gene editing. In the first part (a), Zinc-finger nucleases (ZFNs), transcription activator-like effector nucleases (TALENs), and CRISPR-Cas systems are capable of generating double-strand breaks (DSBs) at precise locations within the genome. Moving on to the second part (b), the introduction of DNA sequences or mutations into the DNA can be achieved by means of homology-directed repair (HDR) or non-homologous end joining (NHEJ) processes with the aid of a donor template. In mammalian cells, CRISPR-induced DSBs are generally mended via NHEJ, which can result in the incorporation of small insertions and/or deletions (indels), leading to gene inactivation due to frameshift mutations. When two DSBs occur on the same chromosome, a substantial segment can be deleted, whereas DSBs on different chromosomes can give rise to chromosomal rearrangements. The abbreviations found in the figure include dsDNA (double-stranded DNA), PAM (protospacer adjacent motif), sgRNA (single-guide RNA), ssDNA (single-stranded DNA), and TALE (transcription activator-like effector). Reprinted from [111].

Inactivation of oncogenes

The main role of CRISPR-based approaches is to inactivate oncogenes by identifying the specific oncogenes that play a vital role in the progression of tumor [110]. Oncogenes are usually linked with mutations or abnormal amplification of genes that result in the over expression of their respective proteins, which leads leading to unregulated growth and proliferation [111].

Researchers create a guide RNA (gRNA) which specifically recognizes and binds to the mutated or amplified region of the oncogene when the target oncogene has been identified [112]. One approach is to inactivate genes that drive tumor growth. For instance, tumor growth in animal models of lymphoma can be reduced by inactivating the MYC oncogene [113]. This strategy depends on the understanding that oncogenes can become overactive due to the mutations of genes in tumor cells, which results in excessive growth of cells and division. By deactivating these oncogenes, it's possible to halt the proliferation of tumor cells [114]. CRISPR-based strategies can also be effectively combined with other treatments of tumor to enhance their effectiveness and achieve better therapeutic results [115]. For example, the CRISPR technique can be used to specifically target the genes that are responsible for drug resistance during chemotherapy, making tumor cells more responsive to chemotherapy drugs. [116]. Additionally, CRISPR can be used to design patient-derived immune cells, such as T cells, to express CARs that enhance their tumor-targeting capabilities in combination with CAR-T cell therapy. Moreover, CRISPR can disable immune checkpoint genes within tumor cells, which results in enhancing the impact of immunotherapies, such as immune checkpoint inhibitors. Another optimistic application is its combination with targeted therapies; by using gene editing to simultaneously disrupt multiple key signaling pathways, CRISPR can help mitigate drug resistance and strengthen the effectiveness of targeted treatments [118]. By employing CRISPR to enhance drug delivery, researchers can modify tumor cells or the tumor microenvironment to improve the penetration of therapeutics, thereby augmenting the impact of various tumor treatments. These examples demonstrate the versatility of CRISPR in synergizing with other tumor therapies and pave the way for more effective and personalized treatment approaches in the fight against

Table 1 CRISPR-based strategies for cancer therapy

Strategy	Mechanism of Action	Advantages	Disadvantages	Preclinical/clinical results	Ref
Inactivating genes that drive tumor growth	Targeting and disrupting oncogenes and tumor suppressor genes to stop cancer cell growth and induce apoptosis	Precision targeting: CRISPR can be designed to target specific genes or mutations that drive tumor growth, increasing specificity and reducing off-target effects. High efficacy: Studies have shown that inactivating certain genes using CRISPR can result in tumor regression and increased survival in animal models	Potential for off-target effects: While CRISPR offers high specificity, there is still the potential for unintended changes in the genome that could cause harm to the patient. Difficulty targeting specific genes or delivering therapy to tumor site: Some tumors may be difficult to target using current delivery methods	Study showed efficacy of using CRISPR to inactivate KRAS oncogene in mouse model of lung cancer. This is a major milestone since KRAS mutations are notoriously difficult to target with other therapies	[16]
Enhancing immune response to cancer cells	Editing immune cells to recognize and destroy cancer cells, such as by editing T cells to express chimeric antigen receptors (CARs)	Enhances body's natural immune response: By editing immune cells to recognize and attack cancer cells, CRISPR-based immunotherapy can activate the body's natural immune response to fight the cancer. Avoids toxic effects of chemotherapy: Unlike chemotherapy, which can have significant side effects, immunotherapy using CRISPR-edited T cells has the potential to be a more targeted and less toxic treatment approach	Potential for toxicity or immune rejection: There is a risk that CRISPR-edited immune cells could attack healthy cells or be rejected by the patient's immune system. Limited availability of specific T cells for editing: It can be challenging to obtain a sufficient number of T cells for editing, which could limit the widespread use of this approach	Study reported complete remission in two out of three patients treated with CRISPR-edited T cells in a trial for refractory lymphomas. This is a promising result, although more research is needed to determine the safety and efficacy of this approach in larger patient populations	[17]
Repairing genetic mutations that cause cancer	Correcting genetic mutations in tumor suppressor genes, DNA repair genes, or other driver genes	Precision targeting: CRISPR can be used to correct specific genetic mutations that cause cancer, potentially leading to long-term benefits. Potential for long-term benefits: Repairing genetic mutations that cause cancer could potentially result in long-term benefits for patients	Potential for off-target effects: As with other CRISPR-based approaches, there is a risk of unintended changes to the genome that could cause harm to the patient. Difficulty delivering therapy to tumor site: It can be challenging to deliver CRISPR-based therapy directly to the tumor site	Promising results in preclinical studies, such as using CRISPR to correct BRCA1 mutations in ovarian cancer cells	[18]
Delivering cancer-killing molecules directly to tumor cells	Using CRISPR to edit the genome of a virus or bacteria to specifically target cancer cells, delivering therapeutic molecules such as toxins or immune modulators directly to tumor cells	Precision targeting: By editing the genome of a virus or bacteria to specifically target cancer cells, CRISPR-based therapy can offer highly targeted and specific delivery of therapeutic molecules to tumor cells. High efficacy: Studies have shown that CRISPR-mediated delivery of cancer-killing molecules can result in tumor regression and increased survival in animal models	Potential for off-target effects: As with other CRISPR-based approaches, there is a risk of unintended changes to the genome that could cause harm to the patient. Limited availability of specific viruses or bacteria for editing: It can be challenging to obtain a sufficient number of specific viruses or bacteria for editing, which could limit the widespread use of this approach	Promising results in preclinical studies, such as using CRISPR to deliver CRISPRa to activate tumor-suppressive microRNAs in liver cancer cells. Other examples include using CRISPR to deliver toxic payloads to tumor cells, such as in a study where CRISPR was used to engineer bacteria to produce a toxin that specifically targets cancer cells in a mouse model of pancreatic cancer	[17, 18]

tumor [119].

Enhancement of immune response

The mechanism centers on using the CRISPR-Cas system, which is originally a natural immune defense present in bacteria and archaea, which has been used for exact gene editing in a wide range of organisms [120]. Researchers use CRISPR-Cas to strengthen the immune response by modifying genes that play vital roles in immune regulation and signaling pathways [121]. To enhance the immune function it includes knocking out genes that overcome immune function [120]. Moreover, CRISPR-based methods enables the accurate insertion of beneficial genes, such as those encoding cytokines or other immune-boosting molecules, to amplify the immune system's response to specific antigens[115]. Additionally, CRISPR-Cas can be deployed to immune cells such as T-cells and NK cells, enhancing their ability to specifically target and attack tumor cells or infected cells [122]. Moreover, CRISPR-Cas enables the development of genetic vaccines, where specific antigen-encoding genes are delivered into host cells to trigger a strong and targeted immune response. This advancement in CRISPR-based immune enhancement offers significant potential to fight against infectious diseases, tumor, and other conditions where strengthening of immune system is required for effective treatment [123]. Researchers have used CRISPR-based gene editing to increase the expression of the PD-1 protein on T cells, which helps to improve their ability to target and kill tumor cells [124]. CRISPR-Cas stimulates the immune response by enabling accurate gene editing [125]. Scientists can target specific genes involved in immune regulation and response pathways. CRISPR-Cas increases the overall activity of the immune system by eliminating the genes that negatively regulate the immune system [115]. Furthermore, the immune response against specific antigens can further enhance by inserting beneficial genes, such as cytokines or other immune mediators, using CRISPR-Cas [126]. CRISPR based gene editing can sometimes result in non-specific site effects, where unexpected changes occur in other parts of the genome. These non-selective site effects may result in unwanted alterations in gene function and could pose safety concerns in the context of immune enhancement [127, 128]. It is important to completely evaluate and minimize these undesirable site effects to ensure the safety and effectiveness of CRISPR-based strategies [120].

Although CRISPR-based approaches carry important potential in strengthening the immune response, it is crucial to assess their long-term impact on the host's immune system [122]. Continuous activation or alteration of immune pathways may result in immune dysregulation, autoimmunity, or immune exhaustion [129].

It is important to understand how CRISPR-mediated immune enhancements affect overall immune function and balance to establish their safe and sustainable use in clinical settings. Long-term follow-up studies in animal models and clinical trials will be necessary to address these concerns. [115]. Figure 7 illustrates various applications of CRISPR in tumor research.

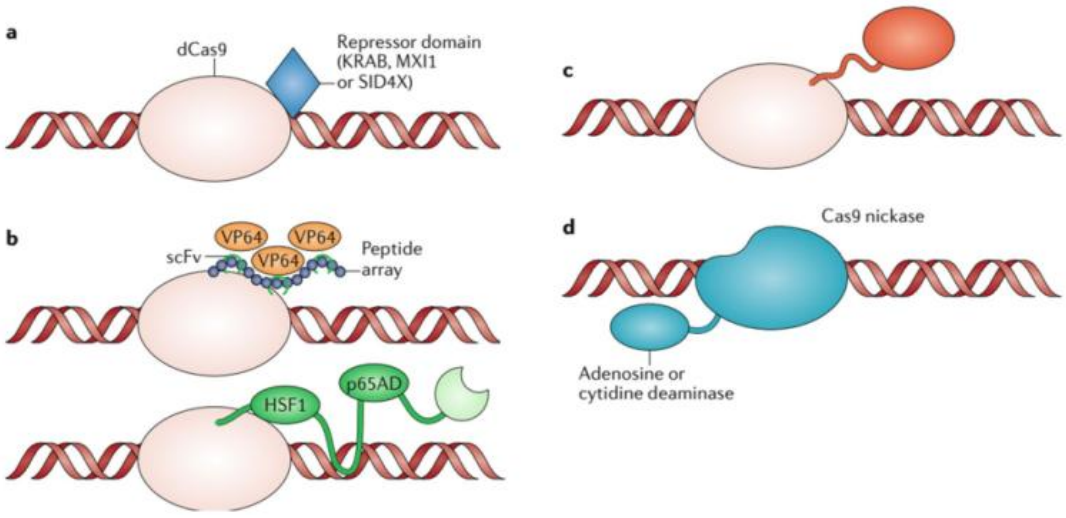


Fig. 7 The diverse applications of CRISPR technology within tumor research. In section a, the paragraph explains that the inhibition of a specific gene can be accomplished by combining Deactivated Cas9 (dCas9) with repressor domains. In section b, it discusses how the fusion of dCas9 with activation domains can stimulate the expression of a particular gene. Furthermore, it mentions that augmenting the binding of additional transcription activators to a single-guide RNA or dCas9 can intensify the expression of target exons. In section c, it states that when dCas9 is fused with epigenetic regulators, it can either repress or activate transcription. In section d, the paragraph talks about the focused introduction of point mutations in the genome, which is made possible by combining dCas9 with adenosine deaminase or cytidine deaminase, allowing for precise genetic modifications. Additionally, it provides explanations for the abbreviations KRAB (Kruppel-associated box) and scFv (single-chain variable fragment). Reprinted from [111] with permission from Springer Nature

Preclinical studies and clinical trials for CRISPR-based tumor therapy

A limited number of CRISPR-based tools have successfully advanced beyond preclinical trials only. However, there have been notable advancements in the CRISPR gene-editing technology, with over 800 cell and gene therapy programs in existence [143]. Other gene editing techniques, such as TALENs and ZFNs, have also been explored commonly in clinical settings and have been reviewed elsewhere. Figure 8 shows different strategies for editing cells using CRISPR technology in patients. The advancement and innovation in CRISPR-based tumor therapy is a speedily evolving and is moving from preclinical studies to clinical trials [146]. Preclinical studies are vital for evaluating the safety and efficacy of CRISPR based tumor therapy before testing directly to the humans [147]. Clinical trials are used to determine the safety and efficacy of a therapy in humans and are the final step in the development process.

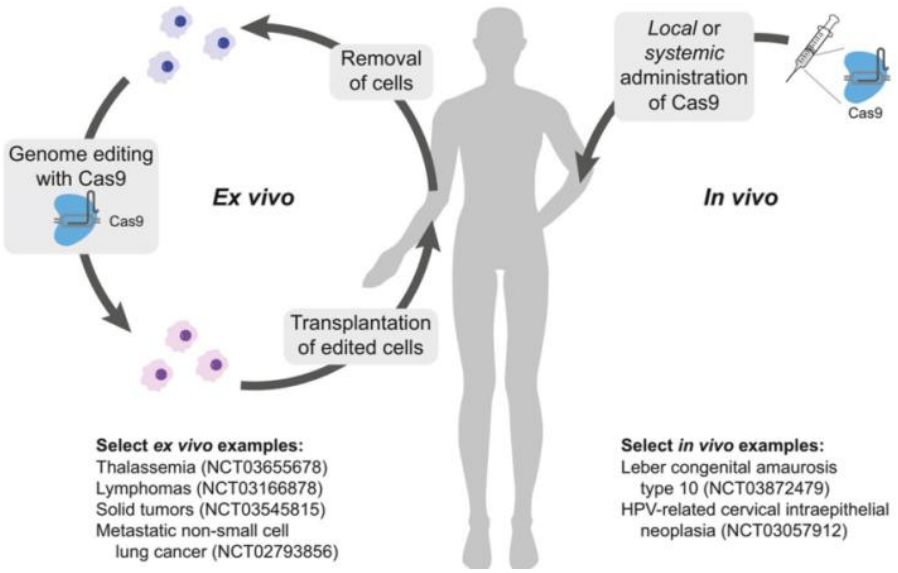


Fig. 8 Different strategies for editing cells using CRISPR technology in patients. On the left, ex vivo applications involve first isolating cells, then expanding along with editing them before transplanting them back. On the right, in vivo editing takes place by the administration of CRISPR-Cas9 (or dCas9, not shown) locally or systemically using viral packaging or nanoparticles. [109]

Preclinical studies

Preclinical studies are a vital step in the advancement of any new tumor therapy, including those techniques that are based on CRISPR technology [148]. Various studies conducted in laboratory animals, such as mice and rats are used to assess the safety, efficiency, and the side effects of a new therapy. Figure 9 illustrates the application of CRISPR in tumor modeling for cells and mice. Martinez-Lage et al. presented a unique and efficient preclinical approach that targets oncogenic gene fusions that targets tumor cells selectivity along with the disruption of a tumor-promoting genetic lesion. This technique tookadvantage of the unique fusion characteristic and describes potential effectiveness [149]. Another preclinical example by Gao et al. focused on exploiting nuclear factor-κB (NF-κB), which is activated in tumor cells, to drive the transcription of CRISPR-Cas13a components which resulted in tumor cell-restricted oncogene silencing. It also offers a promising avenue for tumor therapy [150]. Table 2 shows an overview of preclinical studies exploring the potential of CRISPR-based tumor therapy across many cell types.

The EGFR (Epidermal Growth Factor Receptor) gene plays a vital role in the growth of cell, proliferation, and differentiation [151]. Mutations in the EGFR gene are associated with various tumors, specifically in NSCLC. CRISPR-Cas9 is a gene-editing technology which uses a guide RNA to target particular DNA sequences. It also uses Cas9 enzyme to create double-strand breaks at the targeted location. These breaks can be repaired by NHEJ or HDR and results in gene mutations or precise edits, respectively [152]. In the context of EGFR, CRISPR-Cas9 can be engineered to target and modify the mutated sequences that are responsible for tumor growth,slowing tumor progression and improving patient outcomes potentially [153]. Editing the EGFR gene using CRISPR-Cas9 can have both positive and negative effects [151]. On the positive side, it can help correct mutations in the gene that are linked with particular diseases, such as lung tumor [154]. However,it is important to keep in view potential non-targeted site effects, as unintended changes in other parts of the genome. This can cause unexpected effects or disruptions in function of gene [155].

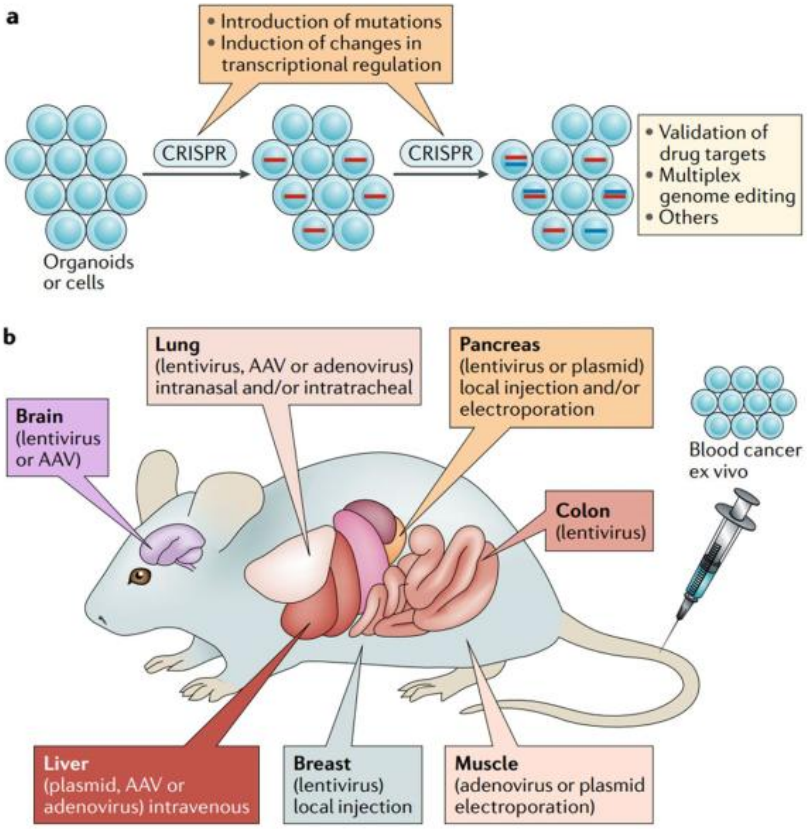


Fig. 9 Using CRISPR for creating tumor models in cells and mice. In the initial case (a), cultured cells or organoids undergo genome editing through CRISPR. It aids in the incorporation of changes and modifications in transcriptional regulation at many phases. In the latter case (b), CRISPR techniques can be delivered to animal models using various methods, allowing them to target numerous tissues and organs. One such technique involves utilizing adeno-associated viruses (AAV) for delivery[111].

The efficacy of CRISPR-Cas9 in editing the EGFR gene can vary depending on various factors such as including the specific gRNA design, delivery method, and cell type [152]. Studies have shown that by using CRISPR-Cas9 high editing efficiency can be attained. But it is important to keep in view the experimental conditions to maximize the expected outcomes [151]. CRISPR-Cas9 has high specificity because of precise binding of the guide RNA to the target DNA sequence [154]. Cas9 may unintentionally edit other genomic regions with a little similarity to the target site so, there is always a risk of non-specific site effects. [151]. Unwanted site effects had been reduce significantly because of extensive optimized research of guide RNA design[154]. State-ofthe-art Cas9 variants, such as high-fidelity Cas9 and base editors, have further improved specificity, lowering the risk of unplanned genetic modifications [151].

Difficulties regarding CRISPR-Cas9 editing of the EGFR gene includes unwanted site effects along with delivery efficiency and long-term effects potentially. Ethical considerations includes informed consent along with ensuring equal access to the technology, and responsible use so that unintended consequences or the creation of "designer babies can be avoided." Laborious assessment, ruling, and ethical guidelines are important to navigate these difficulties andensure the responsible application of CRISPR-Cas9 in editing the EGFR gene or any other gene [155]. CRISPR-Cas9 editing of the VEGF (Vascular Endothelial Growth Factor) gene can play a vital role in various tumors. VEGF is a protein that encourages the growth of new blood vessels, a process known as angiogenesis, which is crucial for tumor development and metastasis[156].

By aiming and upsetting the VEGF gene using CRISPR-Cas9, researchers can potentially hamper the production of VEGF and, resultantly, hinder tumor angiogenesis. This could lead to reduced tumor growth and increased sensitivity to other tumor handlings [157]. Since CRISPR-Cas9 is highly specific, there is a possibility of off-target effects where unplanned gene edits occur. In the case of VEGF gene editing, researchers must carefully assess potential off-target sites to ensure that no critical genes are accidentally modified [158].

To select guide RNAs with the least likelihood of non-specific site activity, laborious bioinformatics analyses and advanced screening methods are used to lower unwanted site effects. CRISPR-Cas9-based VEGF gene editing, on its own, may not be appropriate for complete tumor treatment [156]. While it can hinder tumor angiogenesis, an inclusive tumor treatment technique usually involves combining CRISPR-Cas9 with other therapies like chemotherapy, radiation, or immunotherapy [157]. Combining treatments can resul to a synergistic impact, pointing

tumor cells through various pathways and increasing the overall therapeutic efficiency[156].

Table 2 Preclinical studies of CRISPR-based cancer therapy

Cell/Tissue Type	CRISPR Approach	Results	Limitations/Challenges	Ref
Mouse and Human T cells	Knockout of PD-1	Enhanced T cell activity against cancer cells, prolonged survival in mice	Safety concerns with long-term PD-1 knockout	[59]
Ovarian Cancer Cells	Knockout of oncogene	Reduced cell proliferation and colony formation	Limited assessment of off-target effects	[60]
Lung Cancer Cells	Knockout of mutant EGFR	Reduced cell viability and tumor growth in mice	Off-target effects in some cells	[61]
Triple-Negative Breast Cancer Cells	Knockout of AXL	Reduced tumor growth and increased sensitivity to chemotherapy	Limited assessment of off-target effects	[62]
Various Cancer Cell Lines	Knockout of HIF-1α	Reduced tumor growth and increased sensitivity to radiation therapy	Limited assessment of off-target effects	[63]
Melanoma Cells	Knockout of BRAF V600E	Reduced tumor growth and increased sensitivity to targeted therapy	Off-target effects in some cells	[64]
Leukemia Cells	Knockout of MCL-1	Induced apoptosis and reduced tumor growth	Limited assessment of off-target effects	[65]
Human T cells	Knockout and overexpression of NKG2D	Enhanced tumor cell recognition and killing by CAR-T cells	Safety concerns with long-term NKG2D overexpression	[66, 67]
Pediatric Solid Tumors	Knockout of fusion oncogenes	Reduced tumor growth and increased sensitivity to chemotherapy	Limited assessment of off-target effects	[68]
Prostate Cancer Cells	Knockout of androgen receptor	Reduced cell proliferation and tumor growth in mice	Off-target effects in some cells	[69]
Pancreatic Cancer Cells	Knockout of TGF-β pathway genes	Reduced tumor growth and increased sensitivity to chemotherapy	Limited assessment of off-target effects	[70]
Breast Cancer Cells	Knockout of PAK1 C-terminus	Reduced cell proliferation and tumor growth in mice	Off-target effects in some cells	[71]
Diffuse Large B-cell Lymphoma Cells	Knockout of BCL6	Induced apoptosis and reduced tumor growth	Limited assessment of off-target effects	[72]
Glioblastoma Cells	Knockout of TERT promoter mutations	Reduced cell proliferation and tumor growth in mice	Limited assessment of off-target effects	[73]
Acute Myeloid Leukemia Cells	Knockout of GATA2	Induced differentiation and reduced tumor growth	Limited assessment of off-target effects	[74]
Human Hematopoietic Stem Cells	Knockout of BCL11A enhancer	Induced fetal hemoglobin expression and reduced sickle cell symptoms	Limited assessment of off-target effects	[75]
B Cell Acute Lymphoblastic Leukemia Cells	Knockout of CD19	Induced apoptosis and reduced tumor growth in mice	Off-target effects in some cells	[76]
Diffuse Large B-cell Lymphoma Cells	Knockout of EZH2	Reduced cell proliferation and tumor growth in mice	Limited assessment of off-target effects	[77]
Glioma Cells	Knockout of IDH1	Reduced cell viability and tumor growth in mice	Off-target effects in some cells	[78]
Neuroblastoma Cells	Knockout of MYCN	Induced apoptosis and reduced tumor growth	Limited assessment of off-target effects	[79]
Ovarian Cancer Cells	Knockout of MUC16	Reduced cell proliferation and invasion	Limited assessment of off-target effects	[80]
Alveolar Rhabdomyosarcoma Cells	Knockout of PAX7-FOXO1	Reduced cell proliferation and tumor growth in mice	Off-target effects in some cells	[81]
Esophageal Cancer Cells	Knockout of SOX2	Reduced cell proliferation and colony formation	Limited assessment of off-target effects	[82]
Hepatocellular Carcinoma Cells	Knockout of TERT	Reduced cell proliferation and tumor growth in mice	Limited assessment of off-target effects	[83]
Colorectal Cancer Cells	Knockout of Wnt/β-catenin pathway genes	Reduced cell proliferation and tumor growth in mice	Off-target effects in some cells	[84]

Delivering CRISPR-Cas9 components to tumor cells poses a momentous trial [158]. The large size of the Cas9 protein and guide RNA complex may bound delivery technique [158]. Many methods are being discovered which include viral vectors, nanoparticles, and liposomes, to guarantee effective and safe delivery to target tumor cells while eluding harm to healthy tissues [156]. Ensuringlong-term VEGF gene suppression is necessary for sustained therapeutic effects [158]. Researchers are investigating approaches to improve CRISPR-Cas9 delivery and constancy within tumor cells [156]. Approaches like employing adapted Cas9 variants or integrating the CRISPR components into the genome of the target cells could potentially improve the stability of VEGF gene editing and its antitumor impacts [158]. CRISPR-Cas9 editing of the PTEN gene has noteworthy impact on tumor development [159]. PTEN is a tumor suppressor gene that normalizes cell growth and division [160]. When PTEN is mutated, it leads to uninhibited cell growth, a trademark of tumor [161]. Researchers can potentially restore its function as a tumor suppressor By using CRISPR-Cas9 to precisely target and edit the PTEN gene, thereby hindering tumor cell growth and lowering tumor development rate [162]. The potential of CRISPR-Cas9-mediated PTEN gene editing as a treatment option differs among various tumor types [160]. Some tumors exhibit PTEN mutations as a leading driver of tumorigenesis which makes them more responsive to this attitude [159]. As some tumors may possess alternative mechanisms to bypass PTEN function, the efficiency of this approach depends on the tumor's genetic framework, [162]. General preclinical studies and clinical trials are essential to determine its applicability and efficacy across various tumor types [159]. Safety concerns in CRISPR-Cas9 gene editing for PTEN in tumor therapy includes potential unwanted site effects, where unplanned genetic changes could occur in nontumorous cells which leads to hostile effects [160]. The risk of introducing new mutations or changing other vital genes must be carefully assessed to avoid surplus side effects [162]. Rigorous testing in preclinical models and careful monitoring during clinical trials are essential to ensure the feasibility of this therapeutic method [159]. Researchers are continuously discovering various techniques to improve CRISPR-Cas9 gene editing efficacy [160]. One method includes enhancing the delivery system to ensure accurate targeting of tumor cells. The progressions in CRISPR-Cas9 technology, such as using base editors or prime editors, offer more accurate alterations and reduced non-targeted site effects [162]. Moreover, combining CRISPRCas9 with other therapies, such as immunotherapies or targeted therapies, may improve the overall therapeutic response which allows for a more inclusive and operative treatment strategy. Several difficulties need to be addressed when using CRISPR-Cas9 to edit the PTEN gene in tumor cells [159]. Firstly, effective delivery of CRISPR components to specific tumor cells is vital to avoid non-targeted site impacts [162]. Secondly, ensuring the accurate and precise editing of the PTEN gene without
introducing unplanned mutations is important for therapeutic success [159]. Additionally, the immune response to the CRISPR mechanisms and potential immune rejection of edited cells must be assessed to measure their long-term feasibility and safety [162]. The TERT gene, which encodes the telomerase reverse transcriptase enzyme, plays a vital role in upholding telomeres, the defensive caps at the ends of chromosomes [163]. In many tumor types, the TERT gene is upregulated, leading to increased telomerase activity. This allows tumor cells to avoid the natural drawbacks on cell division and achieve immortality, aiding in tumor growth and development. CRISPR-Cas9 can be employed as a gene-editing tool to target and modify the TERT gene in tumor cells [164]. The CRISPR-Cas9 system consists of a guide RNA that aims at the Cas9 nuclease to the preferred genomic location [165]. Researchers can guide Cas9 to the TERT gene and suggest a DSB atthe pointed site, by designing a guide RNA specific to the TERT gene sequence, [166]. The cell's DNA repair machinery then maintains the break, often through the error-prone NHEJ pathway, which familiarizes small insertions or deletions that interrupt the TERT gene's function[163]. Instead, researchers can use CRISPR-Cas9 along with a repair template to initiate specific changes to the TERT gene sequence, such as gene knockouts or point mutations [165]. The editing of TERT gene using CRISPR-Cas9 can result in several outcomes. The disruption of TERT gene function is one possible result, that leads to decreased telomerase activity in tumor cells. This can result in telomere shortening and cellular senescence or apoptosis,that inhibits the unlimited replicative potential of tumor cells [163]. The modification of TERT gene expression is another possibility, such as reduction of its expression level, which can stop the growth of tumor [166]. Moreover, editing of CRISPR-Cas9-mediated TERT gene may enhance sensitivity of tumor cells to other therapies, as inhibition of telomerase can increase the effectiveness of conventional treatments like chemotherapy or radiotherapy [165].
A crucial protein called Cyclin-Dependent Kinase 4

that is involved in cell cycle regulation is encoded by the CDK4 gene [165]. CDK4 forms complexes with cyclin D, resulting in the progression of cell cycle from G1 to S phase [168].This has been observed in various tumors, that promotes uncontrolled cell proliferation and tumorigenesis [169]. In CRISPR-Cas9, guide RNA molecules are used that bind to the target DNA sequence complementarily within the CDK4 gene [168]. The guide RNA then indicates the precise location where the Cas9 protein cuts the DNA, acting as molecular scissors. The double-strand breaks in the CDK4 gene resulting from this trigger the cell's DNA repair machinery, which may cause gene knockout or targeted mutations [169].Promising results have been shown by preclinical studies using CRISPR-Cas9 in the inhibition of CDK4 expression and reduction of tumor growth in various tumor types of tumors, such as melanoma, breast tumor, and glioblastoma. These studies have provided understanding of the potential therapeutic efficacy of CRISPR-based CDK4 targeting [170].
Currently, several clinical trials are being carried out or planned to evaluate the safety and efficiency of CRISPR-Cas9 in editing the CDK4 gene in tumor patients. These trials will help in determination of the feasibility and potential benefits of CRISPR-based techniques in real-world clinical settings [171].The efficient delivery of CRISPR-Cas9 components to the tumor cells is one challenge. Successful therapy can be carried out by ensuring high delivery rates and minimizing non-targeted site effects [168]. Moreover, targeting it may disrupt other important functions of CDK4 in non-tumorous tissues [169].

Clinical trials

Clinical trials for exploring the potential of CRISPR-based tumor therapy are currently in underway. Multiple clinical trials have been launched to test the safety and effectiveness of CRISPR-based tumor treatment for humans. Frangoul et al.'s research focuses on using CRISPR-Cas9 gene editing to address two severe monogenic diseases [172].

One approach is to knockout the PD1 and the TCR gene (T cell receptor) simultaneously in allogeneic mesothelin-targeting Chimeric Antigen Receptor (CAR) T cells. This technique is being tested in the Phase I clinical trial NCT03545815 [173]. The removal of TCR can cause CAR T cells to avoid adverse effects like graft-versus-host disease (GVHD) while targeting mesothelin expressing tumor cells. Some trials further focus on targeting other genes to improve CAR T cell therapy. For example, the aim of clinical trial NCT04037566 is to improve the effectiveness of CD19-targeting CAR T cells by editing the endogenous HPK1 gene [174]. Similarly, endogenous CD5 knockout is used by NCT04767308 in allogeneic CD5-targeting CAR T cells during the early phase I trial to improve their efficacy [175]. Furthermore, NCT03166878 uses the knockout of both TCR and β2m (beta-2 microglobulin) in allogeneic CD19-targeting CAR T cells [175]. The removal of TCR prevents GVHD, and the elimination of β2m reduces expression of MHC-I, which can otherwise suppress CAR T cell activity. Other tests in Table 2 include the insertion of CAR into T cells after knocking out specific genes. For example, CRISPR is used to insert the CAR gene by NCT04502446 while simultaneously knocking out endogenous TCR and MHC-I in allogeneic CD70-targeting CAR T cells . In a similar way, NCT04244656 targets BCMA (B cell maturation protein) in allogeneic CAR T cells while knocking out endogenous TCR and MHC-I to enhance their tumor-targeting ability NCT04637763 explores the knockout of PD1 and endogenous TCR in allogeneic CD19-targeting CAR T cells [176]. The aim of this approach is to improve the persistence and activity of CAR T cells in attacking CD19-expressing tumor cells. Table 3 shows clinical trials of CRISPR-based tumor therapies targeting different tumor types.

Safety and delivery challenges

CRISPR-Cas9, is highly promising for the treatment of various genetic disorders and diseases [177]. However, like other techniques, its widespread adoption has significant safety and delivery challenges [178]. One primary challenge is its potential to introduce genetic changes that are not intended, known as undesirable site effects, which could trigger new health problems[179]. More precise and reliable methods must be developed by researchers and scientists to minimize these non-targeted site effects for the safety and efficacy of the treatment [180, 181]. Delivering the CRISPR-Cas9 components into specific cells and tissues may also pose a significant hurdle [182, 183]. It is important to find efficient delivery methods that can effectively target the intended cells while avoiding adverse reactions in surrounding tissues [177]. Overcoming these safety and delivery challenges is crucial to unlock the full potential of CRISPR-Cas9 as a transformative therapeutic approach. Robust research and rigorous testing is important to ensure that the benefits of this groundbreaking technology outweigh any potential risks[179]. Table 4 presents the safety and delivery challenges linked with CRISPR-based tumor therapy.

Table 3 Clinical trials of CRISPR-based cancer therapy

Cancer Type	Treatment Approach	Patient Population	Description	Advantages	Disadvantages	Ref
Metastatic melanoma	TCR/CAR-T therapy targeting NY-ESO-1	Patients with NY-ESO-1 + tumors who failed prior therapy	T cells were edited to express NY-ESO-1 TCR/CAR, infused back into patients	Target specificity, long-term persistence	Possible off-target effects, limited efficacy in some patients	[160]
Non-Hodgkin's Lymphoma	CD19-targeting CAR-T therapy	Patients with refractory/relapsed NHL	T cells were edited to express CD19 CAR, infused back into patients	High response rate, durable response in some patients	Cytokine release syndrome, neurotoxicity, potential for tumor antigen escape	[161]
Bladder Cancer	PD-1 knockout via CRISPR/Cas9	Patients with high-risk non-muscle-invasive bladder cancer	CRISPR-edited autologous T cells were infused into patients	Potential for enhanced anti-tumor immune response	Off-target effects, potential for immune-related adverse events	[162]
Sarcoma	Targeted genome editing of PAX3-FOXO1 fusion gene	Patients with metastatic sarcoma expressing PAX3-FOXO1	CRISPR/Cas9 was used to target the fusion gene in tumor cells, followed by infusion of edited T cells	Specific targeting of oncogenic driver mutation	Off-target effects, limited efficacy in some patients	[163]
Solid Tumors	Adoptive transfer of TCR/CAR-T cells targeting neoantigens	Patients with advanced solid tumors	T cells were edited to express TCR/CAR targeting neoantigens unique to each patient's tumor, infused back into patients	Target specificity, potential for durable response	Heterogeneity of tumor neoantigens, potential for off-target effects	[164]
Renal Cell Carcinoma	PD-1 knockout via CRISPR/Cas9	Patients with advanced RCC who failed prior therapies	CRISPR-edited autologous T cells were infused into patients	Potential for enhanced anti-tumor immune response	Off-target effects, potential for immune-related adverse events	[165]
Multiple Myeloma	BCMA-targeting CAR-T therapy	Patients with relapsed/refractory multiple myeloma	T cells were edited to express BCMA CAR, infused back into patients	High response rate, durable response in some patients	Cytokine release syndrome, neurotoxicity	[166]
Glioblastoma	EGFRvIII-targeting CAR-T therapy	Patients with recurrent GBM expressing EGFRvIII	T cells were edited to express EGFRvIII CAR, infused back into patients	Target specificity, potential for durable response	Heterogeneity of tumor antigen expression, potential for off-target effects	[167]
Esophageal Cancer	Targeted genome editing of MUC1 gene	Patients with MUC1 + esophageal squamous cell carcinoma	CRISPR/Cas9 was used to target the MUC1 gene in tumor cells, followed by infusion of edited T cells	Specific targeting of oncogenic driver mutation	Off-target effects, limited efficacy in some patients	[168]
Prostate Cancer	Targeted genome editing of androgen receptor gene	Patients with castration-resistant prostate cancer	CRISPR/Cas9 was used to target the androgen receptor gene in tumor cells, followed by infusion of edited T cells	Specific targeting of oncogenic driver mutation	Off-target effects, limited efficacy in some patients	[169]
Acute Myeloid Leukemia	CD33-targeting CAR-T therapy	Patients with relapsed/refractory AML	T cells were edited to express CD33 CAR, infused back into patients	Target specificity, potential for durable response	Cytokine release syndrome, neurotoxicity	[170]
Head and Neck Cancer	Targeted genome editing of HPV16 E6 gene	Patients with HPV16 + recurrent or metastatic head and neck cancer	CRISPR/Cas9 was used to target the HPV16 E6 gene in tumor cells, followed by infusion of edited T cells	Specific targeting of oncogenic driver mutation	Off-target effects, limited efficacy in some patients	[171]

Table 4 Safety and delivery challenges of CRISPR-based cancer therapy

Challenge	Current Approaches/ Strategies	Limitations	Future Directions	Ethical and Regulatory Considerations	Description	Novelty	Advantages	Disadvantages	Limitations/ Challenges	Ref
Off-Target Effects	Targeted Genome Editing	- Can be difficult to achieve high specificity	- Development of improved delivery methods and monitoring techniques	- Need for clear guidelines and oversight	CRISPR can cause unintended changes to the genome outside the target region, which can lead to adverse effects	CRISPR-mediated genome editing is a highly precise and targeted approach to gene therapy	- Minimizes damage to healthy cells	- Possibility of unintended changes to the genome	Specificity needs to be improved to avoid off-target effects	[206, 207]
Delivery to Tumor Site	Use of Viral Vectors	- Lack of specificity, difficulty targeting tumor cells	- Development of targeted delivery systems	- Need to minimize risk to healthy tissues	Viral vectors are commonly used to deliver CRISPR-based therapies, but can be difficult to target to specific tumor cells	Targeted delivery of CRISPR-based therapies can minimize damage to healthy tissues	- Can be used to target specific tumor cells	- Can cause immune reactions and toxicity	Targeting specificity needs to be improved to avoid damage to healthy tissues	[208, 209]
Safety	Monitoring Off-Target Effects	- Limited sensitivity and accuracy of current monitoring techniques	- Development of more sensitive and accurate monitoring techniques	- Need to ensure patient safety and minimize potential harm	Monitoring of off-target effects is essential to ensure the safety of CRISPR-based therapies	Improved monitoring techniques can provide more accurate information about the effects of CRISPR editing on the genome	- Can help to detect unintended changes to the genome	- Current techniques have limited sensitivity and accuracy	More research is needed to improve monitoring techniques	[210, 211]
Immune Reactions	Use of Non-Immunogenic CRISPR Systems	- Limited availability of non-immunogenic CRISPR systems	- Development of non-immunogenic CRISPR systems	- Need to minimize the risk of immune reactions and toxicity	CRISPR-based therapies can trigger immune responses, which can lead to adverse effects	Non-immunogenic CRISPR systems are being developed to minimize the risk of immune reactions	Non-immunogenic CRISPR systems can reduce the risk of immune reactions and toxicity	- Non-immunogenic CRISPR systems are not widely available	More research is needed to develop non-immunogenic CRISPR systems	[212, 213]

Transport Mechanisms and Challenges in Crossing the Blood-Brain Barrier (BBB)

The brain is shielded by the highly specialized blood-brain barrier (BBB), which strictly controls the movement of essential metabolic molecules between the bloodstream and the brain (Fig. 10) [205,206,207]. The BBB is composed of multiple layers that together form a physical and enzymatic barrier, significantly limiting the entry of various substances [208]. As its core, a layer of specialized brain microvascular endothelial cells (BMECs) serves as the primary barrier, continuously interacting with the surrounding cerebral vasculature to maintain brain homeostasis [205].

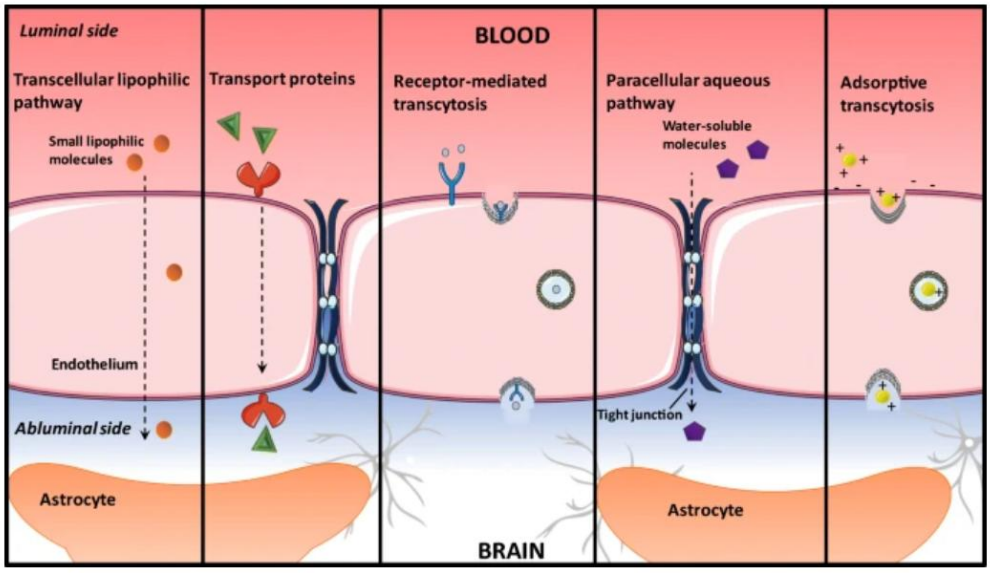


Fig.10 An overview of the transport mechanisms across the blood-brain barrier (BBB) reveals that it is composed of tightly connected endothelial cells lining the brain's capillaries. This structure restricts access to brain tissue while allowing the selective passage of essential nutrients necessary for normal metabolic function. The strict regulation of brain homeostasis prevents many small and large therapeutic agents from passively crossing the BBB through either transcellular or paracellular routes. As a result, energy-dependent transport mechanisms must be employed, including receptor-mediated transcytosis, adsorptive transcytosis, and the use of specialized transport proteins.

Transport across the BBB

The blood-brain barrier (BBB) represents a significant obstacle in the delivery of therapeutics for central nervous system (CNS) disorders [209]. Conventional strategies to overcome this barrier include direct intracerebral injection nad temporary disruption of the BBB to enhance drug penetration. Wu et al. employed microbubbles in combination with focused ultrasound (FUS) to transiently open the BBB via localized cavitation [210]. In their study, microbubbles were administered intravenously to Sprague-Dawley rats, followed by FUS at three different mechanical index (MI) levels: MI=0.62, 0.85, and 1.30. Assessment using Evans blue (EB) staining demonstrated increased BBB permeability with FUS and microbubble application. However, magnetic resonance imaging and hematoxylin and eosin staining revealed signs of brain tissue damage. Notably, EB-stained brain regions were evident at MI levels of 0.85 and 1.38, indicating that FUS intensity must be carefully regulated to ensure safety [210]. An alternative method involves co-administering the therapeutic agent with hyperosmolar mannitol [211], which promotes the rapid movement of fluid from cerebral endothelial cells into the blood vessels, temporarily opening tight junctions [212,213]. Nevertheless, this method carries the potential risk of permanent BBB damage, requiring precise control. Consequently, leveraging the BBB's natural transport systems presents a more favorable and safer route for therapeutic delivery [214].

Safety measures in CRISPR-based gene editing

It is important to evaluate off-target effects in gene editing to ensure the safety and precision of genetic modifications [184]. The aim of advancements in base editors and prime editors is to minimize non-selective site effects, but ongoing research and ethical considerations are necessary to utilize these technologies responsibly for therapeutic applications and other genetic interventions [185]. Off-target effects are the unintended changes or alterations in the DNA caused by using gene editing technologies like CRISPR-Cas systems. These unintended modifications can happen at sites that are not the targeted location, resulting in unpredictable and unwanted genetic changes [184]. A recent

advancement in CRISPR technology are the Base editors that can perform targeted chemical modifications to specific DNA bases without creating double-stranded breaks like traditional CRISPR-Cas systems. This targeted approach reduces the risk of undesirable site effects by reducing random DNA alterations[185]. Compared to base editors or traditional CRISPR-Cas systems, Prime editors offer enhanced precision in gene editing. They combine the capabilities of base editors and nickases, and allow for accurate insertion, deletion, and substitution of particular genetic bases within the genome. This increased precision further reduces the risk of non-targeted site effects [186]. Various techniques such as whole-genome sequencing, high-throughput sequencing, and computational analysis are used by scientists to assess non-selective site effects. These methods help in the identification of unintended genetic changes and determine the efficiency and specificity of the gene-editing technology being used [187]. Off-target effects remain a concern even after the advancements in base editors and prime editors [187]. The challenge is to achieve absolute precision in targeting specific genomic sites without affecting nearby regions [185]. These challenges can be overcome by continuous refinement of gene-editing tools, along with rigorous evaluation and validation methods [186]. Comprehensive preclinical studies are necessary to ensure the safety of using base editors and prime editors in therapeutic settings. Rigorous testing of the gene-editing tools on relevant cell lines and animal models is involved in these studies. The purpose of this is to assess potential non-targeted site effects and ensure the accuracy of genetic modifications before progressing to human trials [185].

High-fidelity Cas9 variants, which have been engineered to have reduced off-target activity are used in one approach. [185]. Using alternative CRISPR systems such as Cpf1 or Cas12a, which have unique mechanisms of target recognition and have shown to have lower off-target activity compared to Cas9 is another approach[188]. One more technique is using a combination of gRNA and Cas9 variants with high specificity, or using multiple gRNAs to target the same gene which increases the specificity of the CRISPR-based gene editing and reduces the risk of non-targeted site effects [189]. Computational tools can also be used to minimize off-target effects in order to predict potential undesirable site sites and to experimentally validate these predictions which helps the researchers to identify and avoid non-selective site sites before they cause unintended mutations [190]. It is important to note that off-target effects can also result from the delivery method that is used to deliver the CRISPR machinery to the cells [191]. Variety of methods are being developed by the researchers to deliver CRISPR to the tumor site, including viral vectors, nanoparticles, and exosomes[192]. Research by Xiang et al. is focused on improving the efficiency prediction of CRISPR-Cas9 gRNAs by using data integration and deep learning (Fig. 11). The main purpose of this research is to improve the accuracy of identifying gRNAs that are more effective in targeting specific DNA sequences. CRISPR-Cas9, which is a powerful gene-editing tool, relies on gRNAs to guide the Cas9 enzyme to the target DNA site for editing. Efficient gRNA design is therefore, important for successful genome editing. This requires reliable predictions of on-target efficiency. The researchers gathered high-quality gRNA activity data for 10,592 gRNAs that target the SpCas9 enzyme to achieve this target. The researchers integrated this new data with existing complementary data from other sources to further improve their predictions. A deep learning model called "CRISPR on," was then employed by them which was trained on a combined dataset of 23,902 gRNAs, including both the newly generated data and the previously data.

The results of their study showed that CRISPR on outperformed existing tools used for gRNA efficiency prediction. The researchers observed improved performances across four test datasets that were not part of the training data used for developing other prediction tools. CRISPR on's predictions were found to be more accurate and reliable than what was currently available. The researchers developed an interactive webserver for gRNA design based on the CRISPR on standalone software to make their findings accessible to the scientific community. The researchers can easily access and use the CRISPR on tool for designing gRNAs with higher efficiency using this webserver [193]. However, more research is needed to optimize the effectiveness and safety of these delivery methods because they are still in early stages of development [194].A major concern in CRISPR-based gene editing are undesirable site effects. These site effects are particularly relevant in the context of tumor therapy[194]. New strategies including using high fidelity Cas9 variants, alternative CRISPR systems, computational tools, and optimized delivery methods are being developed by researchers to minimize the risk of non-selective site effects. [194, 195]. These strategies have shown promise. However much work remains to be done to ensure that CRISPR-based gene editing is safe and effective for tumor therapy [196].

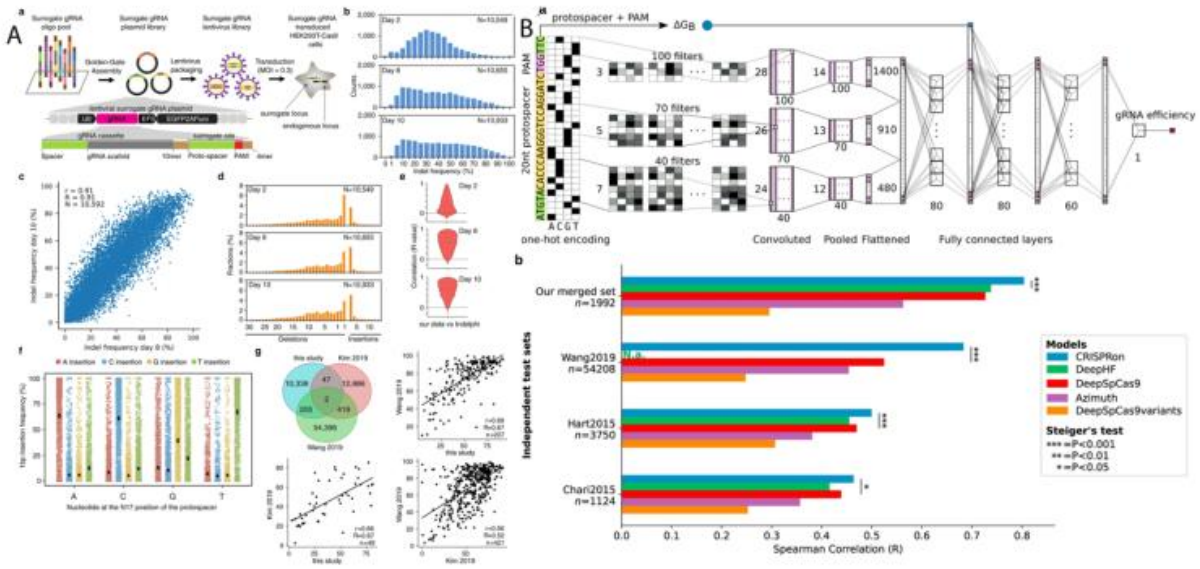


Fig. 11 A The process and results of high-throughput quantification of gRNA efficiency in cells. In panel (a), a graphic illustrates the sequence of actions involved, which includes employing a lentiviral surrogate vector, synthesizing an oligo pool, performing PCR amplification, using golden-gate assembly, packing the genetic material into lentiviruses, and then introducing it. Panel (b) showcases the editing efficiency of gRNA at all surrogate locations, assessed through targeted amplicon sequencing. The results are presented for HEK293T-SpCas9 cells at 2, 8, and 10 days following the introduction. Panel (c) displays the correlation between gRNA editing efficiency on days 8 and 10 post-transduction. Panel (d) presents the patterns of indels (deletions ranging from 1–30 bp and insertions ranging from 1–10 bp) introduced by SpCas9 in HEK293T-SpCas9 cells at 2, 8, and 10 days after the transduction. Panel (e) depicts the agreement between the observed indel patterns in cells and those predicted by inDelphi, visualized as a violin plot with medians and quartiles. In panel (f), a scatter plot portrays the frequency of 1-bp insertion indels (mean ± 95% confidence interval), categorized based on the nucleotide at position N17 of the protospacer and the type of inserted nucleotide. Lastly, panel (g) exhibits the association between gRNA editing efficiencies in this study and those from other significant research, with a particular emphasis on common gRNA + PAM (23 nt) cases, presented using a Venn diagram. **B** The CRISPR on model and its ability to generalize on independent test sets. Panel a displays a visual depiction of the input DNA sequence for CRISPRon, including the prediction algorithm. The deep learning network receives inputs in the form of a one-hot encoded 30mer and the binding energy (ΔGB). It's worth noting that only the filtering (convolutional) layers and the three fully connected layers are explicitly depicted, with the thin vertical bars representing the output of one layer, serving as the input for the next layer. In panel b, a performance evaluation comparing CRISPRon to other existing models is presented, specifically focusing on independent test sets containing over 1000 gRNAs. [193]

The unintended changes in the DNA of cells caused by the CRISPR-Cas9 system are referred to as non targeted site effects.The genes that were not intended to be targeted by the CRISPR-Cas9 system can undergo these changes which can compromise the therapeutic effect of gene editing and potentially lead to undesirable and harmful consequences[197]. The PIK3CA gene in human tumor cells was targeted by the researchers using CRISPR-Cas9. PIK3CA is a well-established oncogene. An oncogene is a gene that promotes the development of tumor. Unintended mutations caused by the CRISPR-Cas9 system were found by the researchers in several non-targeted site genes, including the AKT1 gene. The therapeutic effect of targeting PIK3CA could have been compromised by these undesirable site effects, as these mutations could have activated AKT1 and thus lead to tumor growth[198]. Similarly, the KRAS gene in human tumor cells was targeted by some other researchers using CRISPR-Cas9. KRAS is also a widely recognized oncogene. The findings of investigators showed that CRISPR-Cas9 system caused unintended mutations in several off-target genes, including the NF1 gene. The therapeutic effect of targeting KRAS could have been compromised by these non-selective side effects, as these mutations could have inactivated NF1 and thus enhanced tumor growth [199]. MYC gene in human tumor cells was targeted by the researchers using CRISPRCas9. Like KRAS, MYC is also a well-established oncogene. They found that unintended mutations were caused by the CRISPR-Cas9 system in several off-target genes, including the BCL2L11 gene. The therapeutic effect of targeting MYC could have been compromised by these undesirable site effects, as these mutations could have activated BCL2L11 and thus lead to tumor growth [200]. The TERT gene in human tumor cells was targeted using CRISPR-Cas9 by some other researchers. TERT is another well-established gene that promotes the growth of tumor. They found unintended mutations were caused by the CRISPR-Cas9 system in several off-target genes, including the NFE2L2 gene. The therapeutic effect of targeting TERT could have been compromised by these non-targeted site effects, as these mutations could have inactivated NFE2L2 and thus lead to tumor growth [163].The investigation of the potential therapeutic

effect of targeting the BRCA1/TP53/RAS gene was the main purpose of using CRISPR-Cas9, which is commonly mutated in breast and ovarian tumor/tumor [201]. Unintended mutations were the undesired consequences of using CRISPR-Cas9 on the BRCA1/TP53/RAS gene in several off-target genes, including RAD51D/MDM2/ MAPK, respectively [202] which could have compromised the therapeutic effect of targeting BRCA1/TP53/RAS as they had the ability to activate RAD51D/ inactivate MDM2/activate MAPK, leading to the promotion of tumor growth [203]. In these studies, the frequent findings of unintended mutations in undesirable site genes highlight the significant risk of non-selective site effects linked with CRISPRCas9 gene editing [204].

Conclusion

This scoping review presents a comprehensive analysis of CRISPR Cas 9 gene editing as a groundbreaking approach to addressing the multifaceted challenges of glioblastoma (GBM). By organizing gene targets into specific functional categories, the study underscores the wide-ranging utility of CRISPR Cas 9 in altering critical cellular processes within the GBM microenvironment. The findings question traditional perspectives on gene roles and reveal novel therapeutic opportunities, particularly in targeting molecular regulators of cell cycle progression.

CRISPR Cas 9 driven interventions further refine our understanding of cellular dynamics, shedding light on interphase mechanisms, cell renewal, and migration patterns. The precision of this technology enables researchers to unravel intricate molecular pathways that dictate growth, division, and differentiation at crucial phases of the cell cycle.

Additionally, the review highlights CRISPR Cas 9's potential to decode the complexity of the GBM microenvironment, offering pathways for innovative, targeted therapies.

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